

The random irregular subgraph has Property $(*)$ for all $d = o(n/\log n)$: a proof of the Fox–Luo–Pham conjecture

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Machine-generated note

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Abstract

Let G be a d -regular graph on n vertices. Assign independent uniform labels $x(v) \in [0, 1]$ to the vertices and keep an edge uv of G exactly when $x(u) + x(v) \geq 1$; call the resulting spanning subgraph $H = H(G)$. Fox, Luo and Pham proved that if $d = o(n/(\log n)^{12})$ then with high probability every degree $0 \leq k \leq d$ is attained by $(1 + o(1))n/(d+1)$ vertices of H (Property $(*)$), and conjectured that the same holds for all $d = o(n/\log n)$. We prove this conjecture. Quantitatively, for $n \geq 3$ and $d \geq 64 \log n$, with probability at least $1 - 5n^{-3}$ every degree count is within $6401(\sqrt{(n/d) \log n} + \log n)$ of $n/(d+1)$, uniformly over G ; a separate read- $(d+1)$ Chernoff bound covers $d < 64 \log n$. The proof exposes, for each vertex, the bit $\mathbf{1}\{x(v) \geq \frac{1}{2}\}$: after this exposure every edge inside the lower half is absent, every edge inside the upper half is present, and only the cut edges remain random. The random part of the model is therefore bipartite for *every* G , and a read- d Hölder inequality of Finner together with the $B(1, r)$ law of uniform spacings gives the concentration at the correct scale $\sqrt{n/d}$. The statement concerns the unmodified random subgraph $H(G)$ itself; it is not implied by the recent resolution of the Alon–Wei conjecture, which modifies $H(G)$.

This note was produced by AI within the [Graph Theory AI](#) project: the argument was found by a language model and audited by an adversarial LLM referee, and it has not been checked by a human mathematician. [Section 6](#) records the full provenance.

1 Introduction

Throughout, G is a finite simple undirected d -regular graph on n vertices, $N_G(v)$ is the neighbourhood of v and $N_G[v] = N_G(v) \cup \{v\}$. The *irregular random subgraph* $H = H(G)$, introduced by Frieze, Gould, Karoński and Pfender [2] and studied by Alon and Wei [3] and by Fox, Luo and Pham [1], is defined as follows: assign to every vertex an independent label $X_v \sim \text{Unif}[0, 1]$ and let

$$uv \in E(H) \iff uv \in E(G) \text{ and } X_u + X_v \geq 1.$$

For $0 \leq k \leq d$ write $m_k = m(H, k) = |\{v : \deg_H(v) = k\}|$, and put

$$\mu = \frac{n}{d+1}.$$

A spanning subgraph H of G has *Property $(*)$* if $m(H, k) = (1 + o(1))n/(d+1)$ for all $0 \leq k \leq d$, i.e. if $\max_{0 \leq k \leq d} |m_k - \mu| = o(\mu)$. Since $\mathbb{P}(\deg_H(v) = k) = 1/(d+1)$ exactly for every v and k ([Lemma 4.1\(c\)](#)), $\mathbb{E}m_k = \mu$, and Property $(*)$ asks that all $d+1$ degree counts be simultaneously concentrated at their common mean.

Alon and Wei [3] conjectured that every d -regular graph on n vertices contains a spanning subgraph with $(1 + o(1))n/(d+1)$ vertices of each degree $0 \leq k \leq d$ (Conjecture 1.2 of [1]; their Conjecture 1.1

asks for $n/(d+1) \pm 2$. Fox, Luo and Pham [1, Theorem 1.3] proved that the random subgraph $H(G)$ itself has Property (*) with high probability whenever $d = o(n/(\log n)^{12})$, by a vertex-exposure martingale with a Bernstein-type inequality; one logarithm is lost in the bound on the martingale increments and eleven in the bound on the predictable quadratic variation. They then stated the following, in prose, at the end of their Section 1 (the wording below is that of the problem catalogue this note answers).

Conjecture 1.1 (Fox, Luo and Pham [1, Section 1]). The irregular random subgraph $H = H(G)$ satisfies Property (*) — that is, $m(H, k) = (1 + o(1))n/(d+1)$ for all $0 \leq k \leq d$ — with high probability for all $d = o(n/\log n)$.

The authors of [1] also remark that for $d = \omega(n/\log n)$ the random subgraph $H(G)$ is likely to fail Property (*), so the conjectured range is the natural one. We prove [Conjecture 1.1](#) ([Corollary 2.2](#)), with a quantitative bound in which the deviation is $O(\sqrt{(n/d) \log n} + \log n)$, which is $o(\mu)$ exactly when $d \log n = o(n)$ ([Theorem 2.1](#)).

Concurrent work, and what this note does and does not claim. While the machine-generated argument presented here was being produced and refereed, the motivating Alon–Wei conjecture was settled by other means. Montgomery, Pokrovskiy and Sudakov [4] proved that for $d = o(n)$ every d -regular graph on n vertices contains a spanning subgraph with $(1 + o(1))n/(d+1)$ vertices of each degree, proving Conjecture 1.2 of [1] in full; and Cao, Tang and Wu [5] proved the exact Conjecture 1.1 (within ± 2) for every fixed d and all sufficiently large n . The proof in [4] starts from a typical random subgraph $H(G)$ and then *modifies* it by a correction that redistributes degrees; the properties of $H(G)$ used there are aggregated over blocks of degrees, not per-degree counts. (The referee did not examine the method of [5], and this note makes no claim about it.) Both papers cite $d = o(n/(\log n)^{12})$ from [1] as the state of the art for the *unmodified* random subgraph, and [4] repeats the remark of [1] that $H(G)$ itself is not expected to satisfy (*) in the whole range $d = o(n)$. The present note concerns the unmodified random subgraph $H(G)$ only: it proves that $H(G)$ itself has Property (*) for all $d = o(n/\log n)$, which is the range [1] conjectured and, by their remark, the range in which this is plausible. This statement is neither implied by nor implies the results of [4, 5]; what is new here is the range $o(n/\log n)$ for the random model, and the method ([Section 4](#)). The adversarial referee ([Appendix B](#)) searched the citing literature of [1] and found no paper improving the range for the random model beyond $o(n/(\log n)^{12})$. *[The two references [4, 5] and this paragraph were added at the rewriting stage, following the referee report; the original writeup states that literature priority had not been checked.]*

The idea. Write each label as $X_v = (C_v + U_v)/2$ with $C_v \sim \text{Bernoulli}(1/2)$ and $U_v \sim \text{Unif}[0, 1]$ independent. Once the bits $C = (C_v)$ are exposed, an edge uv of G with $C_u = C_v = 0$ has $X_u + X_v \leq 1$ and is (almost surely) absent from H , an edge with $C_u = C_v = 1$ has $X_u + X_v \geq 1$ and is present, and an edge with $C_u \neq C_v$ is present exactly when $U_u + U_v \geq 1$. Thus the randomness that survives the exposure lives on the cut between $V_0 = \{C_v = 0\}$ and $V_1 = \{C_v = 1\}$, a *bipartite* graph, regardless of the structure or chromatic number of G . Each side of the cut is an independent set of that bipartite graph, so conditioning on the labels of one side makes the degree indicators of the other side independent Bernoulli variables whose success probabilities are uniform spacings; a read- d Hölder inequality (Finner [6]) then gives concentration of m_k around its conditional mean $Q_k(C) = \mathbb{E}[m_k \mid C]$ with variance proxy n/d , and $Q_k(C)$ is in turn a sum of $4/d$ -bounded read- $(d+1)$ functions of independent bits, hence concentrated with the same proxy. Three deviations of order $\sqrt{n/d}$ against a mean $n/(d+1)$ give relative error $o(1)$ precisely when $d \log n = o(n)$, after a union bound over the $d+1$ degrees.

2 Main result

Theorem 2.1 (Main theorem). *Let G be a d -regular graph on $n \geq 3$ vertices and $H = H(G)$ its irregular random subgraph; $\mu = n/(d+1)$.*

(i) *If $d \geq 64 \log n$, then*

$$\mathbb{P}\left(\max_{0 \leq k \leq d} |m_k - \mu| > 6401 \left(\sqrt{\frac{n}{d} \log n} + \log n\right)\right) \leq 5n^{-3}.$$

(ii) *For every $d \geq 0$ and every $0 < \varepsilon \leq 1$,*

$$\mathbb{P}\left(\max_{0 \leq k \leq d} |m_k - \mu| \geq \varepsilon \mu\right) \leq 2(d+1) \exp\left(-\frac{\varepsilon^2 n}{3(d+1)^2}\right).$$

Both bounds are uniform over the choice of G .

Corollary 2.2. *Conjecture 1.1 holds: for every sequence of d -regular graphs G_n on n vertices with $d = d(n) = o(n/\log n)$, the random subgraph $H(G_n)$ satisfies $\max_{0 \leq k \leq d} |m_k - \mu| = o(\mu)$ with probability $1 - o(1)$.*

Proof. Let $d \log n = o(n)$. For those n with $d \geq 64 \log n$, [Theorem 2.1\(i\)](#) applies, and

$$\begin{aligned} \frac{6401(\sqrt{(n/d) \log n} + \log n)}{\mu} &= 6401 \frac{(d+1)\sqrt{(n/d) \log n} + (d+1) \log n}{n} \\ &\leq 6401 \left(2\sqrt{\frac{d \log n}{n}} + \frac{2d \log n}{n}\right) = o(1), \end{aligned}$$

using $(d+1)/\sqrt{d} \leq 2\sqrt{d}$ and $d+1 \leq 2d$ for $d \geq 1$; the failure probability is at most $5n^{-3}$. For those n with $d < 64 \log n$, apply [Theorem 2.1\(ii\)](#) with $\varepsilon_n = 4(d+1)\sqrt{(\log n)/n}$. Since $d+1 \leq 65 \log n$ for $n \geq 3$, $\varepsilon_n \leq 260(\log n)^{3/2}n^{-1/2} = o(1)$, so $\varepsilon_n \leq 1$ for all large n , and then the failure probability is at most $2(d+1) \exp(-\frac{16}{3} \log n) \leq 2n^{-13/3}$. Along the whole sequence the failure probability therefore tends to 0 and the relative deviation tends to 0. (The case $d = 0$ is deterministic, $m_0 = n = \mu$, and is also covered by [\(ii\)](#).) $\square$

Idea of proof (of [Theorem 2.1](#)). Part [\(ii\)](#) is a read- $(d+1)$ Chernoff bound: the indicator $\mathbf{1}\{\deg_H(v) = k\}$ depends on the labels in $N_G[v]$ only, and each label lies in exactly $d+1$ closed neighbourhoods, so Finner's inequality ([Lemma 3.1](#)) bounds the moment generating function of m_k by that of $(d+1) \text{Bin}(n/(d+1), 1/(d+1))$, whose mean is μ ; the standard Poisson-type Chernoff bound follows. Part [\(i\)](#) is the half-interval argument. After exposing the bits C , [Lemma 4.1](#) identifies $\mathbb{E}[m_k | C] = Q_k(C)$ explicitly. On the event $\mathcal{E}$ that every vertex has at least $d/4$ neighbours across the cut (which fails with probability at most $ne^{-d/16}$), [Proposition 4.2](#) gives $\mathbb{P}(|m_k - Q_k| \geq t | C) \leq 2 \exp(-\frac{1}{800} \min\{dt^2/n, t\})$; [Proposition 4.3](#) shows that a truncated version $\bar{Q}_k$ of Q_k , equal to Q_k on $\mathcal{E}$, is sub-Gaussian with $\mathbb{P}(|\bar{Q}_k - \mathbb{E}\bar{Q}_k| \geq t) \leq 2 \exp(-dt^2/(16n))$ and has $|\mathbb{E}\bar{Q}_k - \mu| \leq ne^{-d/16}$. A union bound over the $d+1$ degrees and the choice $t = 3200(\sqrt{(n/d) \log n} + \log n)$ give [\(i\)](#). [\[full proof\]](#)

3 Two classical tools

3.1 Finner's inequality

The following inequality is due to Finner [\[6\]](#); it is also known as the read- R (or “read- k ”) Hölder inequality, see e.g. Pelekis and Ramon [\[7\]](#). We state it in the form used here, for a product probability space.

Lemma 3.1 (Finner [6]). Let $Z_1, \dots, Z_N$ be independent random variables, let $A_1, \dots, A_m \subseteq \{1, \dots, N\}$, and for each i let $f_i \geq 0$ be a measurable function of $(Z_j)_{j \in A_i}$. Let $c_1, \dots, c_m \geq 0$ satisfy

$$\sum_{i: j \in A_i} c_i \leq 1 \quad \text{for every } j \in \{1, \dots, N\}.$$

Then

$$\mathbb{E} \prod_{i=1}^m f_i^{c_i} \leq \prod_{i=1}^m (\mathbb{E} f_i)^{c_i},$$

with the convention that the right-hand side is $+\infty$ if some $\mathbb{E} f_i = \infty$ with $c_i > 0$. In particular, if every coordinate j belongs to at most R of the sets A_i (the family is read- R), then

$$\mathbb{E} \prod_i f_i \leq \prod_i (\mathbb{E} f_i^R)^{1/R}, \quad (1)$$

and for random variables Y_i , each a function of $(Z_j)_{j \in A_i}$ with $\mathbb{E} e^{\lambda R |Y_i|} < \infty$, and every real λ ,

$$\log \mathbb{E} e^{\lambda \sum_i (Y_i - \mathbb{E} Y_i)} \leq \frac{1}{R} \sum_i \log \mathbb{E} e^{\lambda R (Y_i - \mathbb{E} Y_i)}. \quad (2)$$

Repair (attribution). The original writeup stated (1) without attribution and justified it by “apply Hölder with exponent R to those q factors, adding a constant factor when $q < R$ ”. No constant factor is needed or permitted: over a probability space the missing exponent mass is carried by the constant function 1, whose expectation is 1, and a factor larger than 1 would invalidate the later uses of (1). The inequality itself is correct and is Finner’s [6]. The wrong sketch has been dropped; the standard proof, by integrating one coordinate at a time with the generalized Hölder inequality, is reproduced in [Appendix A](#) for completeness. The referee also checked (1) numerically on 4000 random instances on small product spaces (no violations, worst ratio LHS/RHS = 1.000000).

Idea of proof. Integrate out Z_1 first. The factors that involve Z_1 have exponents summing to at most 1, so the generalized Hölder inequality (padded with the constant function 1) bounds their Z_1 -expectation by $\prod (\mathbb{E}_{Z_1} f_i)^{c_i}$, a product of functions of the remaining coordinates which again satisfies the hypothesis. Induct on N . For (1) take $c_i = 1/R$ and apply the result to f_i^R ; for (2) take $f_i = e^{\lambda(Y_i - \mathbb{E} Y_i)}$. [\[full proof\]](#)

3.2 Uniform spacings

Lemma 3.2. Let $r \geq 1$, let $T_1, \dots, T_r$ be independent $\text{Unif}[0, 1]$ variables with order statistics $T_{(1)} \leq \dots \leq T_{(r)}$, and put $T_{(0)} = 0, T_{(r+1)} = 1$. The $r+1$ spacings $S_j = T_{(j+1)} - T_{(j)}, 0 \leq j \leq r$ (including the two end spacings) are exchangeable, each has the $B(1, r)$ law with density $r(1-x)^{r-1}$ on $[0, 1]$, and for every integer $m \geq 1$

$$\mathbb{E} S_j^m = \frac{m!}{(r+1)(r+2) \dots (r+m)}, \quad \text{in particular } \mathbb{E} S_j = \frac{1}{r+1}. \quad (3)$$

Consequently, if $d \geq 1$ and $r \geq d/4$, then for every real θ with $|\theta| \leq 1/8$,

$$\log \mathbb{E} e^{\theta d S_j} \leq \theta \frac{d}{r+1} + 32\theta^2. \quad (4)$$

The bound (4) also holds, with 0 in place of $d/(r+1)$, for the variable identically equal to 0.

Idea of proof. The spacing vector is uniform on the simplex, so $\mathbb{P}(S_j > x) = (1-x)^r$; the moments follow from the Beta integral. For (4), $r \geq d/4$ gives $\mathbb{E}(d S_j)^m \leq 4^m m!$, so the terms of order $m \geq 2$ in $\mathbb{E} e^{\theta d S_j} - 1$ are bounded by $\sum_{m \geq 2} (4|\theta|)^m \leq 32\theta^2$, and $\log(1+x) \leq x$. [\[full proof\]](#)

The referee verified (3) numerically for $r \leq 29, m \leq 5$ (maximum error $1.5 \cdot 10^{-11}$) and (4) by the exact series $\mathbb{E} e^{s S_r} = \sum_{m \geq 0} s^m / \prod_{i \leq m} (r+i)$ in 60-digit arithmetic for $d \in \{4, \dots, 10^5\}$, all admissible r and 81 values of $\theta \in [-1/8, 1/8]$: no violations, with equality only at $\theta = 0$.

4 The half-interval decomposition

Generate the labels as

$$X_v = \frac{C_v + U_v}{2}, \quad C_v \sim \text{Bernoulli}(1/2), \quad U_v \sim \text{Unif}[0, 1],$$

all $2n$ variables independent; then $X_v \sim \text{Unif}[0, 1]$ and $C_v = \mathbf{1}\{X_v \geq 1/2\}$ almost surely. Put $V_a = \{v : C_v = a\}$ for $a \in \{0, 1\}$, and for each vertex define its *cut degree* and *deterministic offset*

$$r_v = |\{u \in N_G(v) : C_u \neq C_v\}|, \quad b_v = \begin{cases} 0, & C_v = 0, \\ d - r_v, & C_v = 1. \end{cases}$$

For a bit configuration C and $0 \leq k \leq d$ let

$$q_{v,k}(C) = \frac{\mathbf{1}\{b_v \leq k \leq b_v + r_v\}}{r_v + 1}, \quad Q_k(C) = \sum_v q_{v,k}(C), \quad (5)$$

and let $\mathcal{E} = \{r_v \geq d/4 \text{ for every } v\}$.

Lemma 4.1. *With the notation above:*

(a) *Almost surely, for every vertex v ,*

$$\deg_H(v) = b_v + \sum_{\substack{u \in N_G(v) \\ C_u \neq C_v}} \mathbf{1}\{U_u + U_v \geq 1\}. \quad (6)$$

In particular, given C , the degree of $v \in V_a$ is a function of U_v and of $(U_u)_{u \in V_{1-a}}$ only.

(b) $\mathbb{P}(\deg_H(v) = k \mid C) = q_{v,k}(C)$ for every v and k ; hence $\mathbb{E}[m_k \mid C] = Q_k(C)$.

(c) $\mathbb{P}(\deg_H(v) = k) = 1/(d+1)$ for every v and $0 \leq k \leq d$; hence $\mathbb{E}m_k = \mathbb{E}Q_k = \mu$.

(d) *Conditional on $C_v, r_v \sim \text{Bin}(d, 1/2)$; hence $\mathbb{P}(r_v < d/4) \leq e^{-d/16}$ and $\mathbb{P}(\mathcal{E}^c) \leq ne^{-d/16}$.*

Idea of proof. If $C_u = C_v = 0$ then $X_u + X_v = (U_u + U_v)/2 \leq 1$ with equality only on a null set; if $C_u = C_v = 1$ then $X_u + X_v \geq 1$; if $C_u \neq C_v$ then $X_u + X_v \geq 1$ iff $U_u + U_v \geq 1$. This is (a). Given C and $U_v = u$, the sum in (6) is $\text{Bin}(r_v, u)$, and $\int_0^1 \binom{r}{j} u^j (1-u)^{r-j} du = 1/(r+1)$ gives (b); the same integral with $r = d$ (conditioning on X_v instead) gives (c). For (d), the neighbours' bits are independent of C_v and each differs from it with probability $1/2$; apply the multiplicative Chernoff bound. [\[full proof\]](#)

Proposition 4.2 (Concentration given the bits). *Let $d \geq 1$, let $\lambda_0 = \log(9/8)$ and $\lambda_1 = \lambda_0/2$. For every bit configuration $C \in \mathcal{E}$ and every $0 \leq k \leq d$:*

(a) *for $a \in \{0, 1\}$, with $M_{a,k} = \sum_{v \in V_a} \mathbf{1}\{\deg_H(v) = k\}$ and $Q_{a,k} = \sum_{v \in V_a} q_{v,k}$,*

$$\log \mathbb{E}[e^{\lambda(M_{a,k} - Q_{a,k})} \mid C] \leq 100 \frac{|V_a|}{d} \lambda^2 \quad (|\lambda| \leq \lambda_0);$$

(b) $\log \mathbb{E}[e^{\lambda(m_k - Q_k)} \mid C] \leq 200 \frac{n}{d} \lambda^2$ for $|\lambda| \leq \lambda_1$;

(c) *for every $t \geq 0$,*

$$\mathbb{P}(|m_k - Q_k| \geq t \mid C) \leq 2 \exp\left(-\frac{1}{800} \min\left\{\frac{dt^2}{n}, t\right\}\right).$$

Repair (constants). The original writeup states (a)–(c) with unspecified absolute constants $C_0, \lambda_0, C_1, \lambda_1, c$, justified by “ $e^\lambda - 1 - \lambda = O(\lambda^2)$ ”. The values above are one admissible choice, obtained as follows. The bound (4) requires $|\eta| \leq 1/8$ for $\eta = e^\lambda - 1$, which holds iff $\log(7/8) \leq \lambda \leq \log(9/8)$, so $\lambda_0 = \log(9/8) \approx 0.1178$ suffices. On $|\lambda| \leq 1$ one has $0 \leq e^\lambda - 1 - \lambda \leq \lambda^2$ and $\eta^2 \leq 3\lambda^2$, and on $\mathcal{E}$ one has $Q_{a,k} \leq 4|V_a|/d$, whence $C_0 = 4 + 32 \cdot 3 = 100$. The Cauchy–Schwarz step doubles λ , so $\lambda_1 = \lambda_0/2 \approx 0.0589$ and $C_1 = 200$. The Chernoff step gives exponent $\min\{dt^2/(800n), \lambda_1 t/2\}$, and $\lambda_1/2 \approx 0.029 \geq 1/800$. These agree with the values found by the referee ($\lambda_0 \approx 0.117$, $\lambda_1 \approx 0.058$, $C_0 \approx 100$).

Idea of proof. Fix $C \in \mathcal{E}$ and condition further on $(U_u)_{u \in V_{1-a}}$. By Lemma 4.1(a) the indicators $\mathbf{1}\{\deg_H(v) = k\}$, $v \in V_a$, are then independent, and the success probability of v is the $(k - b_v)$ -th spacing $p_{v,k}$ of the r_v points $(1 - U_u)_{u \in N_G(v) \cap V_{1-a}}$ (or 0 if $k - b_v \notin [0, r_v]$). With $\eta = e^\lambda - 1$, $\mathbb{E}[e^{\lambda M_{a,k}} | C] = \mathbb{E}[\prod_v (1 + \eta p_{v,k}) | C] \leq \mathbb{E}[\exp(\eta \sum_v p_{v,k}) | C]$. Each U_u with $u \in V_{1-a}$ enters at most d of the $p_{v,k}$, so (1) with $R = d$ and then (4) with $r = r_v \geq d/4$ give $\log \mathbb{E}[e^{\lambda M_{a,k}} | C] \leq \eta Q_{a,k} + 32|V_a|\eta^2/d$; subtracting $\lambda Q_{a,k}$ and using the elementary bounds in the box gives (a). Cauchy–Schwarz across the two sides (which are not independent) gives (b), and optimizing the Chernoff bound gives (c). *[full proof]*

Proposition 4.3 (Concentration of the conditional mean). *Let $d \geq 1$ and define the truncations $\bar{q}_{v,k} = q_{v,k} \mathbf{1}\{r_v \geq d/4\}$ and $\bar{Q}_k = \sum_v \bar{q}_{v,k}$. Then for every $0 \leq k \leq d$:*

- (a) $0 \leq \bar{q}_{v,k} \leq 4/d$, and $\bar{q}_{v,k}$ is a function of $(C_u)_{u \in N_G[v]}$; each bit C_u enters exactly $d + 1$ of these functions;
- (b) $\mathbb{P}(|\bar{Q}_k - \mathbb{E}\bar{Q}_k| \geq t) \leq 2 \exp(-dt^2/(16n))$ for every $t \geq 0$;
- (c) $0 \leq \mu - \mathbb{E}\bar{Q}_k \leq ne^{-d/16}$;
- (d) $\bar{Q}_k = Q_k$ on the event $\mathcal{E}$.

Idea of proof. On $\{r_v \geq d/4\}$, $q_{v,k} \leq 1/(r_v + 1) < 4/d$; the truncation kills the other branch. The bits are independent, so (2) with $R = d + 1$ and Hoeffding’s lemma [8] for a variable in an interval of length $4/d$ give $\log \mathbb{E}e^{\lambda(\bar{Q}_k - \mathbb{E}\bar{Q}_k)} \leq 2n(d + 1)\lambda^2/d^2 \leq 4n\lambda^2/d$, hence (b). For (c), $\mu - \mathbb{E}\bar{Q}_k = \sum_v \mathbb{E}[q_{v,k} \mathbf{1}\{r_v < d/4\}]$ by Lemma 4.1(c), and $0 \leq q_{v,k} \leq 1$ with Lemma 4.1(d). *[full proof]*

It is important that the bits are *not* conditioned on $\mathcal{E}$ in Proposition 4.3: conditioning would destroy their independence. The truncation achieves the same effect, and (d) transfers the conclusion back to $\mathcal{E}$.

5 Remarks

Remark 5.1 (Scope and interpretation). The theorem is exactly the statement conjectured in [1]: it concerns the unmodified random subgraph $H(G)$, all $d + 1$ degree values simultaneously, with high probability, uniformly over all d -regular graphs G on n vertices. Nothing beyond d -regularity is used: G may be disconnected, and no assumption on its (fractional) chromatic number, girth or symmetry is made. Graphs are finite, simple and undirected. The null event $X_u + X_v = 1$ is ignored. The endpoint degrees $k = 0$ and $k = d$ are covered because Lemma 3.2 includes the two end spacings, and the case $k - b_v \notin [0, r_v]$ (success probability identically 0) is covered by the last sentence of Lemma 3.2. The quantitative form Theorem 2.1(i) is stronger than the conjecture: its deviation $O(\sqrt{(n/d) \log n} + \log n)$ is $o(\mu)$ exactly when $d \log n = o(n)$, so the argument saturates at the conjectured threshold and says nothing about $d = \Omega(n/\log n)$, where [1] expect $(*)$ to fail.

Remark 5.2 (Two roles of $ne^{-d/16}$). In the proof of Theorem 2.1(i) the quantity $ne^{-d/16}$ bounds two different things: the bias $\beta = \max_k |\mathbb{E}\bar{Q}_k - \mu|$, which is a number of vertices and enters the deviation, and the probability $\pi = \mathbb{P}(\mathcal{E}^c)$ of the bad event, which enters the failure probability. The original writeup used one symbol B for both. We keep them apart as β and π in Appendix A, inequality (7); both happen to be at most $ne^{-d/16} \leq n^{-3}$ when $d \geq 64 \log n$.

Remark 5.3 (Why the argument is short). Fox, Luo and Pham [1] use a vertex-exposure martingale; their interval-indicator conditioning appears only locally, in the bound on the martingale increments (their Lemma 3.2), and the eleven extra logarithms come from their bound on the predictable quadratic variation (their Lemma 3.3). The heuristic scale of the fluctuations is n/d in variance, and each of the three pieces $m_k - Q_k$, $\overline{Q}_k - \mathbb{E}\overline{Q}_k$, $\mathbb{E}\overline{Q}_k - \mu$ above is controlled at exactly that scale. The single move that makes this possible is global rather than local: exposing $\mathbf{1}\{X_v \geq 1/2\}$ for all v at once makes the surviving randomness bipartite. The referee records (Appendix B, Caveat 8) that it set out to break the argument and could not, and that it regards the conditional-independence step of Proposition 4.2 as “structurally verified by brute force and, on reflection, simply true”. No human mathematician has checked this.

Remark 5.4 (Relation to the earlier machine attempt). This problem was attacked twice (Section 6). The first attempt, by gpt-5.6-sol, proved Conjecture 1.1 under the additional hypothesis $\chi_f(G) d \log n = o(n)$, in particular for bipartite regular graphs, by decomposing $V(G)$ into independent sets via a fractional colouring; it also proved $\text{Var}(m_k) = O(n/d)$ for every G and identified the missing ingredient as an exponential tail for the residual $R_k = \sum_v (\mathbf{1}\{\deg_H(v) = k\} - p_{v,k})$, which it could not obtain for graphs such as disjoint unions of cliques. The present argument circumvents R_k entirely: after exposing the bits, the random part of H is bipartite for every G , and the loss of the factor $\chi_f(G)$ disappears. The two tools of Section 3 appear in both attempts; they are classical and are attributed here to Finner [6] and to the theory of uniform spacings.

Remark 5.5 (Computational checks by the referee). The referee re-derived every step and ran the scripts in `verification_astra/scripts/2207.13651__00/` (Appendix B). Beyond the checks of Lemmas 3.1 and 3.2 quoted above: the identity (6) was brute-forced on 3148 vertices over 300 random regular graphs ($d \leq 6$, $n \leq 16$) with 0 violations; $q_{v,k} = \mathbb{P}(\deg_H(v) = k \mid C)$ was checked by quadrature for $d \in \{3, 6, 9\}$, both bits, all r and k , with maximum error $1.9 \cdot 10^{-11}$; $\mathbb{E}_C q_{v,k} = 1/(d+1)$ was checked exactly for $d \in \{1, 2, 3, 5, 8, 13, 20, 33\}$ (error $1.4 \cdot 10^{-17}$); the conditional independence in Proposition 4.2 was checked structurally (re-randomizing one U_w , $w \in V_a$, never changed $\deg_H(v)$ for another $v \in V_a$; 0 violations over 300 graphs); the tail Lemma 4.1(d) was checked against the exact binomial distribution for $d \leq 3000$ with 0 violations. On the adversarial family $30 \times K_{10}$ ($n = 300$, $d = 9$; 3000 samples, $k = 4$) the simulated $\text{Var}(m_k - Q_k \mid C) = 32.64$ against the predicted scale $n/d = 33.3$ (a match to 2%), $\mathbb{E}[m_k - Q_k] = -0.05$, and $\text{Var}(Q_k) \in [12.7, 16.5]$ against the proxy $8n/d = 267$ from Proposition 4.3; on $21 \times K_{20}$, $\text{Var}(Q_k) \in [4.8, 7.6]$ against 177; on $K_{100,100}$, $\text{Var}(Q_k) \leq 0.195$ against 16. These are consistency checks of the scales, not proofs.

Remark 5.6 (Novelty). To the best of the referee’s search (Appendix B, “Priority check”), no paper improves the range of [1, Theorem 1.3] for the random model $H(G)$ itself; the 2026 papers [4, 5] that settle the Alon–Wei conjectures by modifying $H(G)$ both cite $o(n/(\log n)^{12})$ as the state of the art for $H(G)$. The novelty check is only as good as what could be found online, and the campaign that produced this note lost priority on other results by margins of days to weeks. The importance of the result is also narrower than when [1] posed the conjecture, since its original motivation, Conjecture 1.2 of [1], is now a theorem [4].

6 Provenance

The mathematics in this note was produced without human intervention, in three stages.

(1) The argument was found by the language model gpt-6-astra (reasoning effort *max*, single autonomous pass, flex service tier) on 2026-09-17, as part of a sweep over open problems catalogued from arXiv (catalog id 2207.13651__00, leg `attacks_retry`; original writeup in `attacks_retry/2207.13651__00/output.md`). This was a *second* attempt at the problem: the first, by gpt-5.6-sol (`attacks/2207.13651__00`, self-reported verdict *partial*), was supplied verbatim in the prompt as an unverified lead, with the instruction to treat it as a lead and not an authority. Its content is summarised in Remark 5.4. The second writeup reuses the shape of the first (read- R Hölder bounds and uniform

spacings) and re-states both tools from scratch; the half-interval decomposition of [Section 4](#) is the new step. Nothing inherited from the first attempt is presented here as new.

(2) An independent adversarial referee agent (claude-opus-5) reviewed the writeup on 2026-09-17. It read the source paper [1] and confirmed the statement of the conjecture, its Theorem 1.3 and the source of the twelve logarithms; re-derived every numbered step of the writeup; verified every identity and inequality numerically ([Section 5](#)); searched the literature for prior or concurrent resolutions and found [4, 5], neither of which implies the result; and returned the verdict MINOR_GAPS: every step VALID except the justification of the read- R Hölder lemma, which is Finner’s inequality asserted with a non-proof and no attribution. It also noted the unspecified constants and the double use of $B = ne^{-d/16}$. Its report is reproduced verbatim in [Appendix B](#).

(3) On 2026-09-17 the writeup was rewritten into the present note by claude-fable-5-1 (final pass 2026-09-18). The text was reorganised, with statements and proof ideas in the main text and full proofs in [Appendix A](#). Every deviation from the original writeup is listed here.

- The read- R product Hölder inequality (the writeup’s Section 2.1) is now stated as Finner’s inequality [6] in its general exponent form, cited, and its incorrect proof sketch (“adding a constant factor when $q < R$ ”) removed; a standard proof is included in [Appendix A](#), and [7] is cited as a secondary reference. (Repair named by the referee.)
- The constants in the writeup’s inequalities (13)–(15), left as absolute $C_0, \lambda_0, C_1, \lambda_1, c$, are instantiated: $\lambda_0 = \log(9/8)$, $C_0 = 100$, $\lambda_1 = \lambda_0/2$, $C_1 = 200$, $c = 1/800$; consequently the writeup’s bound (1), stated with an unspecified C “for all sufficiently large n ”, becomes [Theorem 2.1\(i\)](#) with $A = 3200$, deviation constant 6401 and failure probability $5n^{-3}$, for all $n \geq 3$. The elementary bounds used ($e^\lambda - 1 - \lambda \leq \lambda^2$ and $(e^\lambda - 1)^2 \leq 3\lambda^2$ on $|\lambda| \leq 1$) were checked numerically at the rewriting stage. (Repair named by the referee.)
- The quantity $ne^{-d/16}$, used in the writeup’s inequality (21) both as a deviation and as a probability under the single name B , is split into a bias bound β and a probability bound π ([Remark 5.2](#), inequality (7)). (Repair named by the referee.)
- The introduction cites the concurrent work [4, 5] found by the referee, states precisely that this note concerns the unmodified random subgraph, and says what remains new; the references [2, 3] for the origin of the model and of the conjecture were added from the source paper’s own introduction. (Repair named by the referee.)
- The small-degree bound (the writeup’s Section 7, inequality (24)) is stated for all $d \geq 0$ as [Theorem 2.1\(ii\)](#), and its “usual elementary Chernoff calculation” is spelled out in [Appendix A](#) via comparison with a Poisson moment generating function; the writeup applied it only for $d < 64 \log n$, which is how [Corollary 2.2](#) uses it.
- The general inequality (7) is stated for all $d \geq 1$, as in the writeup, before specialising to $d \geq 64 \log n$.

No other mathematical content was changed. **No human mathematician has checked this argument.** We circulate it for human review, not as a claim to a resolution.

A Full proofs

Lemma 3.1 (Finner [6]). *Let $Z_1, \dots, Z_N$ be independent random variables, let $A_1, \dots, A_m \subseteq \{1, \dots, N\}$, and for each i let $f_i \geq 0$ be a measurable function of $(Z_j)_{j \in A_i}$. Let $c_1, \dots, c_m \geq 0$ satisfy*

$$\sum_{i: j \in A_i} c_i \leq 1 \quad \text{for every } j \in \{1, \dots, N\}.$$

Then

$$\mathbb{E} \prod_{i=1}^m f_i^{c_i} \leq \prod_{i=1}^m (\mathbb{E} f_i)^{c_i},$$

with the convention that the right-hand side is $+\infty$ if some $\mathbb{E}f_i = \infty$ with $c_i > 0$. In particular, if every coordinate j belongs to at most R of the sets A_i (the family is read- R), then

$$\mathbb{E} \prod_i f_i \leq \prod_i (\mathbb{E} f_i^R)^{1/R}, \quad (1)$$

and for random variables Y_i , each a function of $(Z_j)_{j \in A_i}$ with $\mathbb{E} e^{\lambda R |Y_i|} < \infty$, and every real λ ,

$$\log \mathbb{E} e^{\lambda \sum_i (Y_i - \mathbb{E} Y_i)} \leq \frac{1}{R} \sum_i \log \mathbb{E} e^{\lambda R (Y_i - \mathbb{E} Y_i)}. \quad (2)$$

Proof. We use the generalized Hölder inequality: if $h_1, \dots, h_m \geq 0$ are random variables and $\theta_1, \dots, \theta_m \geq 0$ with $\sum_i \theta_i = 1$, then $\mathbb{E} \prod_i h_i^{\theta_i} \leq \prod_i (\mathbb{E} h_i)^{\theta_i}$ (with the stated convention for infinite expectations). We prove the lemma by induction on N . If $N = 0$ every f_i is a constant and there is nothing to prove.

Let $N \geq 1$ and write $\mathbb{E} = \mathbb{E}' \mathbb{E}_1$, where $\mathbb{E}_1$ integrates over Z_1 and $\mathbb{E}'$ over $(Z_2, \dots, Z_N)$; by independence and Tonelli's theorem this is legitimate for nonnegative integrands. Let $J = \{i : 1 \in A_i\}$. For fixed values of $Z_2, \dots, Z_N$, the variables f_i with $i \in J$ are nonnegative functions of Z_1 , and $\sum_{i \in J} c_i \leq 1$ by hypothesis. Apply the generalized Hölder inequality to the family $(f_i)_{i \in J}$ with exponents $(c_i)_{i \in J}$, augmented by the constant function 1 with exponent $1 - \sum_{i \in J} c_i \geq 0$ (its expectation is 1, because the law of Z_1 is a probability measure):

$$\mathbb{E}_1 \prod_{i \in J} f_i^{c_i} \leq \prod_{i \in J} (\mathbb{E}_1 f_i)^{c_i}.$$

Define $g_i = \mathbb{E}_1 f_i$ for $i \in J$ and $g_i = f_i$ for $i \notin J$. Each g_i is a nonnegative measurable function of $(Z_j)_{j \in A_i \setminus \{1\}}$, and the exponent condition $\sum_{i: j \in A_i \setminus \{1\}} c_i \leq 1$ holds for every $j \geq 2$. Since the factors with $i \notin J$ do not depend on Z_1 ,

$$\mathbb{E} \prod_i f_i^{c_i} = \mathbb{E}' \left[\prod_{i \notin J} f_i^{c_i} \cdot \mathbb{E}_1 \prod_{i \in J} f_i^{c_i} \right] \leq \mathbb{E}' \prod_i g_i^{c_i} \leq \prod_i (\mathbb{E}' g_i)^{c_i} = \prod_i (\mathbb{E} f_i)^{c_i},$$

where the second inequality is the induction hypothesis applied to the $N - 1$ coordinates $Z_2, \dots, Z_N$, and the last equality is $\mathbb{E}' \mathbb{E}_1 f_i = \mathbb{E} f_i$ for $i \in J$ and $\mathbb{E}' f_i = \mathbb{E} f_i$ for $i \notin J$.

For (1), apply the lemma to the functions f_i^R with all $c_i = 1/R$; the hypothesis $\sum_{i: j \in A_i} 1/R \leq 1$ is the read- R property. For (2), apply (1) to $f_i = e^{\lambda(Y_i - \mathbb{E} Y_i)} > 0$ and take logarithms. $\square$

Lemma 3.2. Let $r \geq 1$, let $T_1, \dots, T_r$ be independent $\text{Unif}[0, 1]$ variables with order statistics $T_{(1)} \leq \dots \leq T_{(r)}$, and put $T_{(0)} = 0, T_{(r+1)} = 1$. The $r + 1$ spacings $S_j = T_{(j+1)} - T_{(j)}, 0 \leq j \leq r$ (including the two end spacings) are exchangeable, each has the $B(1, r)$ law with density $r(1 - x)^{r-1}$ on $[0, 1]$, and for every integer $m \geq 1$

$$\mathbb{E} S_j^m = \frac{m!}{(r+1)(r+2) \dots (r+m)}, \quad \text{in particular } \mathbb{E} S_j = \frac{1}{r+1}. \quad (3)$$

Consequently, if $d \geq 1$ and $r \geq d/4$, then for every real θ with $|\theta| \leq 1/8$,

$$\log \mathbb{E} e^{\theta d S_j} \leq \theta \frac{d}{r+1} + 32\theta^2. \quad (4)$$

The bound (4) also holds, with 0 in place of $d/(r+1)$, for the variable identically equal to 0.

Proof. The vector $(S_0, \dots, S_r)$ has the uniform distribution (density $r!$) on the simplex $\{s \in [0, 1]^{r+1} : \sum_j s_j = 1\}$, which is invariant under permutations of the coordinates; hence the spacings are

exchangeable and it suffices to treat $S_0 = T_{(1)}$. Now $\mathbb{P}(S_0 > x) = \mathbb{P}(T_i > x \forall i) = (1 - x)^r$, so S_0 has density $r(1 - x)^{r-1}$ on $[0, 1]$, which is $B(1, r)$, and

$$\mathbb{E}S_0^m = \int_0^1 x^m r(1 - x)^{r-1} dx = r B(m + 1, r) = \frac{r m! (r - 1)!}{(r + m)!} = \frac{m!}{(r + 1) \cdots (r + m)}.$$

For (4), let $r \geq d/4$, so $r + 1 > d/4$ and $d/(r + 1) < 4$. By (3), $\mathbb{E}(dS_j)^m = d^m m! / ((r + 1) \cdots (r + m)) \leq m! (d/(r + 1))^m \leq 4^m m!$. Since S_j is bounded, $\mathbb{E}e^{\theta dS_j} = \sum_{m \geq 0} \theta^m \mathbb{E}(dS_j)^m / m!$ with absolutely convergent series for $|\theta| < 1/4$, and therefore, using $\log(1 + x) \leq x$,

$$\begin{aligned} \log \mathbb{E}e^{\theta dS_j} &\leq \mathbb{E}e^{\theta dS_j} - 1 = \theta \frac{d}{r + 1} + \sum_{m \geq 2} \frac{\theta^m \mathbb{E}(dS_j)^m}{m!} \leq \theta \frac{d}{r + 1} + \sum_{m \geq 2} (4|\theta|)^m \\ &= \theta \frac{d}{r + 1} + \frac{16\theta^2}{1 - 4|\theta|} \leq \theta \frac{d}{r + 1} + 32\theta^2 \end{aligned}$$

for $|\theta| \leq 1/8$. For the variable identically 0 the left side is $0 \leq 32\theta^2$. $\square$

Lemma 4.1. *With the notation above:*

(a) *Almost surely, for every vertex v ,*

$$\deg_H(v) = b_v + \sum_{\substack{u \in N_G(v) \\ C_u \neq C_v}} \mathbf{1}\{U_u + U_v \geq 1\}. \quad (6)$$

In particular, given C , the degree of $v \in V_a$ is a function of U_v and of $(U_u)_{u \in V_{1-a}}$ only.

(b) $\mathbb{P}(\deg_H(v) = k \mid C) = q_{v,k}(C)$ for every v and k ; hence $\mathbb{E}[m_k \mid C] = Q_k(C)$.

(c) $\mathbb{P}(\deg_H(v) = k) = 1/(d + 1)$ for every v and $0 \leq k \leq d$; hence $\mathbb{E}m_k = \mathbb{E}Q_k = \mu$.

(d) *Conditional on C_v , $r_v \sim \text{Bin}(d, 1/2)$; hence $\mathbb{P}(r_v < d/4) \leq e^{-d/16}$ and $\mathbb{P}(\mathcal{E}^c) \leq ne^{-d/16}$.*

Proof. Since C_v is a fair bit and U_v is uniform on $[0, 1]$, $(C_v + U_v)/2$ is uniform on $[0, 1]$ and $\mathbf{1}\{X_v \geq 1/2\} = C_v$ unless $U_v = 1$, a null event. Let $uv \in E(G)$. If $C_u = C_v = 0$ then $X_u + X_v = (U_u + U_v)/2 \leq 1$, with equality only if $U_u = U_v = 1$, so $uv \notin E(H)$ almost surely. If $C_u = C_v = 1$ then $X_u + X_v = 1 + (U_u + U_v)/2 \geq 1$ and $uv \in E(H)$. If $C_u \neq C_v$ then $X_u + X_v = (1 + U_u + U_v)/2 \geq 1$ iff $U_u + U_v \geq 1$. Hence, almost surely, a vertex v with $C_v = 0$ receives no edges from its $d - r_v$ neighbours in V_0 , a vertex with $C_v = 1$ receives all $d - r_v$ edges from its neighbours in V_1 , and every vertex receives from each cross neighbour u the edge uv exactly when $U_u + U_v \geq 1$. This is (6), and the final sentence of (a) is read off from it.

(b): Fix C and condition further on $U_v = u$. The r_v indicators $\mathbf{1}\{U_w + u \geq 1\}$, $w \in N_G(v)$ with $C_w \neq C_v$, are independent Bernoulli(u) variables, so $\deg_H(v) - b_v \sim \text{Bin}(r_v, u)$. Integrating over $u \in [0, 1]$, for $0 \leq j \leq r_v$,

$$\mathbb{P}(\deg_H(v) - b_v = j \mid C) = \int_0^1 \binom{r_v}{j} u^j (1 - u)^{r_v - j} du = \binom{r_v}{j} B(j + 1, r_v - j + 1) = \frac{1}{r_v + 1},$$

and the probability is 0 for $j \notin [0, r_v]$. Thus $\mathbb{P}(\deg_H(v) = k \mid C) = q_{v,k}(C)$; summing over v gives $\mathbb{E}[m_k \mid C] = Q_k(C)$.

(c): Condition on $X_v = x$. Each of the d edges uv is retained iff $X_u \geq 1 - x$, independently with probability x , so $\deg_H(v) \sim \text{Bin}(d, x)$ and $\mathbb{P}(\deg_H(v) = k) = \int_0^1 \binom{d}{k} x^k (1 - x)^{d - k} dx = 1/(d + 1)$ for $0 \leq k \leq d$. Summing over v , $\mathbb{E}m_k = \mu$, and $\mathbb{E}Q_k = \mathbb{E}\mathbb{E}[m_k \mid C] = \mu$.

(d): Conditional on C_v , the d bits C_u , $u \in N_G(v)$, are independent fair bits, each different from C_v with probability $1/2$; so $r_v \sim \text{Bin}(d, 1/2)$, with mean $d/2$. The multiplicative Chernoff bound $\mathbb{P}(X \leq (1 - \delta)\mathbb{E}X) \leq \exp(-\delta^2 \mathbb{E}X/2)$ with $\delta = 1/2$ gives $\mathbb{P}(r_v < d/4) \leq \mathbb{P}(r_v \leq d/4) \leq e^{-d/16}$. A union bound over the n vertices gives $\mathbb{P}(\mathcal{E}^c) \leq ne^{-d/16}$. $\square$

Proof of Proposition 4.2. Fix $C \in \mathcal{E}$, k and $a \in \{0, 1\}$. All probabilities and expectations below are conditional on C ; the randomness is in $(U_v)_{v \in V}$, which are independent $\text{Unif}[0, 1]$ variables also conditionally on C .

Conditional independence. Condition additionally on $(U_u)_{u \in V_{1-a}}$. By Lemma 4.1(a), for $v \in V_a$,

$$\deg_H(v) = b_v + |\{u \in N_G(v) \cap V_{1-a} : 1 - U_u \leq U_v\}|,$$

a function of U_v and of the fixed values $(U_u)_{u \in V_{1-a}}$. As the U_v , $v \in V_a$, are independent, the indicators $\mathbf{1}\{\deg_H(v) = k\}$, $v \in V_a$, are conditionally independent Bernoulli variables. Let $p_{v,k}$ be the conditional success probability of v . Writing $T_u = 1 - U_u$ for the r_v points $u \in N_G(v) \cap V_{1-a}$, the set of $x \in [0, 1]$ with exactly j of the T_u at most x is the interval $[T_{(j)}, T_{(j+1)})$, of length the j -th spacing S_j in the notation of Lemma 3.2. Hence

$$p_{v,k} = \begin{cases} S_{k-b_v}((T_u)_{u \in N_G(v) \cap V_{1-a}}), & 0 \leq k - b_v \leq r_v, \\ 0, & \text{otherwise.} \end{cases}$$

Conditionally on C , the T_u are independent uniforms, so in the first case $p_{v,k} \sim B(1, r_v)$ and $\mathbb{E}[p_{v,k} \mid C] = 1/(r_v + 1) = q_{v,k}$, by Lemma 3.2; in the second case $p_{v,k} = 0 = q_{v,k}$. Note that $r_v \geq d/4 \geq 1/4$, so $r_v \geq 1$ and Lemma 3.2 applies.

One side. Let $\eta = e^\lambda - 1 > -1$. By conditional independence and $1 + x \leq e^x$,

$$\mathbb{E}[e^{\lambda M_{a,k}} \mid C] = \mathbb{E}\left[\prod_{v \in V_a} (1 + \eta p_{v,k}) \mid C\right] \leq \mathbb{E}\left[\exp\left(\eta \sum_{v \in V_a} p_{v,k}\right) \mid C\right],$$

valid for either sign of λ since every factor $1 + \eta p_{v,k}$ is positive. The function $f_v = e^{\eta p_{v,k}} > 0$ depends only on $(U_u)_{u \in N_G(v) \cap V_{1-a}}$, and a given U_u , $u \in V_{1-a}$, enters f_v only for $v \in N_G(u) \cap V_a$, i.e. for at most d values of v . Finner's inequality (1) with $R = d$, followed by Lemma 3.2, (4), with $\theta = \eta$ and $r = r_v \geq d/4$ (or its trivial version when $p_{v,k} \equiv 0$), gives for $|\eta| \leq 1/8$

$$\log \mathbb{E}[e^{\lambda M_{a,k}} \mid C] \leq \frac{1}{d} \sum_{v \in V_a} \log \mathbb{E}[e^{\eta d p_{v,k}} \mid C] \leq \frac{1}{d} \sum_{v \in V_a} (\eta d q_{v,k} + 32\eta^2) = \eta Q_{a,k} + 32 \frac{|V_a|}{d} \eta^2.$$

Subtracting $\lambda Q_{a,k}$,

$$\log \mathbb{E}[e^{\lambda(M_{a,k} - Q_{a,k})} \mid C] \leq (e^\lambda - 1 - \lambda) Q_{a,k} + 32 \frac{|V_a|}{d} \eta^2.$$

Now let $|\lambda| \leq \lambda_0 = \log(9/8)$. Then $\eta = e^\lambda - 1 \in [8/9 - 1, 9/8 - 1] = [-1/9, 1/8]$, so $|\eta| \leq 1/8$ as required. Moreover, for $|\lambda| \leq 1$: $0 \leq e^\lambda - 1 - \lambda \leq \lambda^2$ (the ratio $(e^\lambda - 1 - \lambda)/\lambda^2$ is increasing in λ and equals $e - 2 < 1$ at $\lambda = 1$), and $\eta^2 \leq 3\lambda^2$ (for $0 \leq \lambda \leq 1$, $\eta \leq (e - 1)\lambda$ and $(e - 1)^2 < 3$; for $\lambda < 0$, $|\eta| \leq |\lambda|$). Finally, on $\mathcal{E}$, $q_{v,k} \leq 1/(r_v + 1) < 4/d$, so $Q_{a,k} \leq 4|V_a|/d$. Therefore

$$\log \mathbb{E}[e^{\lambda(M_{a,k} - Q_{a,k})} \mid C] \leq \lambda^2 \frac{4|V_a|}{d} + 32 \cdot 3\lambda^2 \frac{|V_a|}{d} = 100 \frac{|V_a|}{d} \lambda^2,$$

which is (a).

Both sides. The variables $M_{0,k} - Q_{0,k}$ and $M_{1,k} - Q_{1,k}$ are not independent given C . By the Cauchy–Schwarz inequality and (a) at 2λ , for $|\lambda| \leq \lambda_1 = \lambda_0/2$,

$$\mathbb{E}[e^{\lambda(m_k - Q_k)} \mid C] \leq \prod_{a \in \{0,1\}} \mathbb{E}[e^{2\lambda(M_{a,k} - Q_{a,k})} \mid C]^{1/2} \leq \exp\left(\frac{1}{2} \cdot 100 \frac{|V_0| + |V_1|}{d} (2\lambda)^2\right) = \exp\left(200 \frac{n}{d} \lambda^2\right),$$

which is (b).

Chernoff bound. Put $Z = m_k - Q_k$ and $V = 200n/d$, so that $\log \mathbb{E}[e^{\lambda Z} \mid C] \leq V\lambda^2$ for $|\lambda| \leq \lambda_1$. For $t \geq 0$ and $0 < \lambda \leq \lambda_1$, Markov's inequality gives $\mathbb{P}(Z \geq t \mid C) \leq \exp(-\lambda t + V\lambda^2)$. If $t \leq 2V\lambda_1$,

choose $\lambda = t/(2V) \leq \lambda_1$ to get $\exp(-t^2/(4V)) = \exp(-dt^2/(800n))$. If $t > 2V\lambda_1$, choose $\lambda = \lambda_1$ to get $\exp(-\lambda_1 t + V\lambda_1^2) \leq \exp(-\lambda_1 t + \lambda_1 t/2) = \exp(-\lambda_1 t/2)$. In both cases

$$\mathbb{P}(Z \geq t \mid C) \leq \exp\left(-\min\left\{\frac{dt^2}{800n}, \frac{\lambda_1}{2}t\right\}\right) \leq \exp\left(-\frac{1}{800} \min\left\{\frac{dt^2}{n}, t\right\}\right),$$

because $\lambda_1/2 = \frac{1}{4} \log(9/8) \approx 0.0294 \geq 1/800$. The same bound holds for $-Z$, and (c) follows. $\square$

Proof of Proposition 4.3. (a): On $\{r_v \geq d/4\}$ we have $r_v + 1 > d/4$, so $0 \leq q_{v,k} \leq 1/(r_v + 1) < 4/d$; on the complement $\bar{q}_{v,k} = 0$. The quantities r_v, b_v and $\mathbf{1}\{r_v \geq d/4\}$ are functions of C_v and $(C_u)_{u \in N_G(v)}$, hence so is $\bar{q}_{v,k}$. A bit C_u enters $\bar{q}_{v,k}$ only if $u \in N_G[v]$, i.e. $v \in N_G[u]$, which happens for exactly $|N_G[u]| = d + 1$ vertices v .

(b): The bits $(C_u)_{u \in V}$ are independent and, by (a), the family $(\bar{q}_{v,k})_v$ is read- $(d + 1)$. Hoeffding's lemma [8] states that a random variable Y with values in an interval of length ℓ satisfies $\log \mathbb{E} e^{s(Y - \mathbb{E}Y)} \leq s^2 \ell^2 / 8$ for all real s . By (2) with $R = d + 1$ and Hoeffding's lemma with $\ell = 4/d$, $s = \lambda(d + 1)$, for every real λ ,

$$\log \mathbb{E} e^{\lambda(\bar{Q}_k - \mathbb{E}\bar{Q}_k)} \leq \frac{1}{d+1} \sum_v \frac{\lambda^2(d+1)^2}{8} \left(\frac{4}{d}\right)^2 = \frac{2n(d+1)}{d^2} \lambda^2 \leq \frac{4n}{d} \lambda^2,$$

using $d + 1 \leq 2d$. Markov's inequality with $\lambda = dt/(8n)$ gives $\mathbb{P}(\bar{Q}_k - \mathbb{E}\bar{Q}_k \geq t) \leq \exp(-\lambda t + 4n\lambda^2/d) = \exp(-dt^2/(16n))$, and likewise for the lower tail.

(c): By Lemma 4.1(c), $\mu = \mathbb{E}Q_k = \sum_v \mathbb{E}q_{v,k}$, so $\mu - \mathbb{E}\bar{Q}_k = \sum_v \mathbb{E}[q_{v,k} \mathbf{1}\{r_v < d/4\}]$. Each term is between 0 and $\mathbb{P}(r_v < d/4) \leq e^{-d/16}$, since $0 \leq q_{v,k} \leq 1$ and by Lemma 4.1(d).

(d): On $\mathcal{E}$ every truncation indicator equals 1. $\square$

Theorem 2.1 (Main theorem). *Let G be a d -regular graph on $n \geq 3$ vertices and $H = H(G)$ its irregular random subgraph; $\mu = n/(d + 1)$.*

(i) *If $d \geq 64 \log n$, then*

$$\mathbb{P}\left(\max_{0 \leq k \leq d} |m_k - \mu| > 6401 \left(\sqrt{\frac{n}{d} \log n} + \log n\right)\right) \leq 5n^{-3}.$$

(ii) *For every $d \geq 0$ and every $0 < \varepsilon \leq 1$,*

$$\mathbb{P}\left(\max_{0 \leq k \leq d} |m_k - \mu| \geq \varepsilon \mu\right) \leq 2(d + 1) \exp\left(-\frac{\varepsilon^2 n}{3(d + 1)^2}\right).$$

Both bounds are uniform over the choice of G .

Proof. Part (i): the master inequality. Let $d \geq 1$ and put

$$\beta = \max_{0 \leq k \leq d} |\mathbb{E}\bar{Q}_k - \mu| \leq ne^{-d/16}, \quad \pi = \mathbb{P}(\mathcal{E}^c) \leq ne^{-d/16},$$

by Proposition 4.3(c) and Lemma 4.1(d) (cf. Remark 5.2). On $\mathcal{E}$, by Proposition 4.3(d), for every k

$$|m_k - \mu| \leq |m_k - Q_k| + |\bar{Q}_k - \mathbb{E}\bar{Q}_k| + |\mathbb{E}\bar{Q}_k - \mu| \leq |m_k - Q_k| + |\bar{Q}_k - \mathbb{E}\bar{Q}_k| + \beta.$$

Hence, for every $t \geq 0$, using Proposition 4.2(c) (averaged over $C \in \mathcal{E}$) and Proposition 4.3(b) and a union bound over the $d + 1$ values of k ,

$$\begin{aligned} \mathbb{P}\left(\max_k |m_k - \mu| \geq 2t + \beta\right) &\leq \pi + \sum_{k=0}^d \mathbb{P}(\mathcal{E} \cap \{|m_k - Q_k| \geq t\}) + \sum_{k=0}^d \mathbb{P}(|\bar{Q}_k - \mathbb{E}\bar{Q}_k| \geq t) \\ &\leq \pi + (d + 1) \left[2 \exp\left(-\frac{1}{800} \min\left\{\frac{dt^2}{n}, t\right\}\right) + 2 \exp\left(-\frac{dt^2}{16n}\right) \right] \\ &\leq \pi + 4(d + 1) \exp\left(-\frac{1}{800} \min\left\{\frac{dt^2}{n}, t\right\}\right), \end{aligned} \tag{7}$$

since $\exp(-dt^2/(16n)) \leq \exp(-dt^2/(800n)) \leq \exp(-\frac{1}{800} \min\{dt^2/n, t\})$. The bad event $\mathcal{E}^c$, common to all k , is counted once. Inequality (7) holds for every $d \geq 1$ and every d -regular G .

Part (i): specialisation. Let $n \geq 3$ and $d \geq 64 \log n$. Then $ne^{-d/16} \leq n \cdot n^{-4} = n^{-3}$, so $\beta \leq n^{-3}$ and $\pi \leq n^{-3}$. Take $A = 3200$ and $t = A(\sqrt{(n/d) \log n} + \log n)$. Then $dt^2/n \geq A^2 \log n \geq A \log n$ and $t \geq A \log n$, so $\exp(-\frac{1}{800} \min\{dt^2/n, t\}) \leq n^{-A/800} = n^{-4}$, and, as $d + 1 \leq n$, (7) gives

$$\mathbb{P}\left(\max_k |m_k - \mu| \geq 2t + \beta\right) \leq n^{-3} + 4n \cdot n^{-4} = 5n^{-3}.$$

Finally $2t + \beta \leq 2t + n^{-3} \leq (2A + 1)(\sqrt{(n/d) \log n} + \log n) = 6401(\sqrt{(n/d) \log n} + \log n)$, because $\log n \geq 1 > n^{-3}$ for $n \geq 3$. This proves (i).

Part (ii). Fix k and let $I_v = \mathbf{1}\{\deg_H(v) = k\}$, so $m_k = \sum_v I_v$. The indicator I_v is a function of the labels $(X_u)_{u \in N_G[v]}$, and each label X_u enters I_v for exactly the $d + 1$ vertices $v \in N_G[u]$; so the family $(I_v)_v$ is read- R with $R = d + 1$, with respect to the independent labels. By Lemma 4.1(c), $\mathbb{E}I_v = p := 1/(d + 1)$. Finner's inequality (1) gives, for every real λ ,

$$\mathbb{E} e^{\lambda m_k} = \mathbb{E} \prod_v e^{\lambda I_v} \leq \prod_v (\mathbb{E} e^{\lambda R I_v})^{1/R} = (1 - p + p e^{\lambda R})^{n/R} \leq \exp\left(\frac{np}{R}(e^{\lambda R} - 1)\right) = \exp(\nu(e^s - 1)),$$

where $s = \lambda R$ and $\nu = np/R = n/(d + 1)^2$, using $1 + x \leq e^x$. Thus $W = m_k/R$ satisfies $\mathbb{E} e^{sW} \leq \exp(\nu(e^s - 1))$ for all real s , the moment generating function of a Poisson variable with mean ν ; note $\mathbb{E}W = \mu/R = \nu$. The standard Chernoff computation for this bound gives, for $0 < \varepsilon \leq 1$: with $s = \log(1 + \varepsilon) > 0$,

$$\mathbb{P}(W \geq (1 + \varepsilon)\nu) \leq \exp(\nu(e^s - 1) - s(1 + \varepsilon)\nu) = \exp(-\nu[(1 + \varepsilon) \log(1 + \varepsilon) - \varepsilon]) \leq e^{-\nu\varepsilon^2/3},$$

and with $s = \log(1 - \varepsilon) < 0$ (for $\varepsilon < 1$; the case $\varepsilon = 1$ follows by letting $s \rightarrow -\infty$),

$$\mathbb{P}(W \leq (1 - \varepsilon)\nu) \leq \exp(\nu(e^s - 1) - s(1 - \varepsilon)\nu) = \exp(-\nu[\varepsilon + (1 - \varepsilon) \log(1 - \varepsilon)]) \leq e^{-\nu\varepsilon^2/2},$$

by the elementary inequalities $(1 + \varepsilon) \log(1 + \varepsilon) - \varepsilon \geq \varepsilon^2/3$ and $\varepsilon + (1 - \varepsilon) \log(1 - \varepsilon) \geq \varepsilon^2/2$ on $(0, 1]$ (for $\varepsilon = 1$ the second reads $1 \geq 1/2$). Since $\{|m_k - \mu| \geq \varepsilon\mu\} = \{W \geq (1 + \varepsilon)\nu\} \cup \{W \leq (1 - \varepsilon)\nu\}$,

$$\mathbb{P}(|m_k - \mu| \geq \varepsilon\mu) \leq 2 \exp\left(-\frac{\varepsilon^2 n}{3(d + 1)^2}\right),$$

and a union bound over the $d + 1$ values of k gives (ii). (For $d = 0$ the bound is trivially true, as $m_0 = n = \mu$.) $\square$

B Report of the LLM referee (verbatim)

The following is the unedited report of the adversarial referee agent (claude-opus-5, 2026-09-17) on the *original* writeup. Its step and equation numbers refer to that writeup, not to this note. The scripts it mentions are in `verification_astra/scripts/2207.13651__00/`. Treat it as machine output, not as an authority.

B.1 Interpretation

The target is the informal conjecture stated in prose at the end of Section 1 of Fox–Luo–Pham, *On random irregular subgraphs* (arXiv:2207.13651, RSA 2024). I pulled the paper and confirmed the sentence verbatim:

“We conjecture that the irregular random subgraph H satisfies Property $\star$ with high probability for all $d = o(n/\log n)$.”

Property (*) there is $m(H, k) \leq (1+o(1)) \ln(d+1)$ for all $0 \leq k \leq d$, in the model $x(v) \sim U[0, 1]$ i.i.d., $uv \in E(H) \iff x(u)+x(v) \geq 1$, G d -regular on n vertices.

The writeup's Theorem is exactly this: for every sequence of d -regular graphs with $d = o(n/\log n)$, w.h.p. $\max_{0 \leq k \leq d} |m_k - \mu| = o(\mu)$, $\mu = n/(d+1)$, with estimates uniform in G . No strengthening, no weakening, no strawman. The quantitative form (1), $\max_k |m_k - \mu| \leq C(\sqrt{(n/d) \log n} + \log n)$ with failure probability $\leq Cn^{-3}$ for $d \geq 64 \log n$, is strictly stronger than what is asked and is the natural sharp form (the deviation is $o(\mu)$ precisely when $d \log n = o(n)$, i.e. exactly at the conjectured threshold). Interpretation is fair.

This is a `attacks_retry` target. The prompt contains an earlier `gpt-5.6-sol` attempt (verdict `partial`) which proved the conjecture only under $\chi_f(G) \leq d \log n = o(n)$. Nothing is inherited uncritically: the new writeup reuses only the *shape* of the earlier argument (read- R Hölder + uniform spacings) and re-states both tools from scratch in §2. Its novel content is §3 (the half-interval bit decomposition), which replaces the fractional colouring and removes the $\chi_f(G)$ loss. I checked §2's two tools independently rather than crediting them from the earlier attempt. The earlier attempt's own diagnosis of what was missing (an exponential tail for the residual $R_k = \sum_v \{1_{\deg=k} - p_{\{v, k\}}\}$ for high-chromatic G) is genuinely circumvented, not re-committed: the new argument never needs R_k at all, because after exposing the bits the *random* part of the graph is bipartite for free.

B.2 Step-by-step findings

Numbering follows the writeup's own equation numbers.

step	label	note
Thm / (1), §1	VALID	Statement is the conjecture; (1) is derived in §6.
(2) Lemma, §2.1 (read- R product Hölder)	GAP (repairable)	The inequality is true — it is Finner's inequality (Ann. Probab. 20 (1992) 1893–1901): if $\sum_{i: j \in A_i} c_i \leq 1$ for every coordinate j then $E \prod f_i^{c_i} \leq \prod (E f_i^{c_i})$; take $c_i = 1/R$ and replace f_i by f_i^R . But the writeup gives no citation and its one-paragraph “proof” is not one: “apply Hölder with exponent R to those q factors, adding a constant factor when $q < R$ ” is wrong-headed — over a <i>probability</i> space no constant is needed (the missing mass sits on the function 1, of norm 1), and a constant > 1 would break the inequality as later used. Verified numerically (0 violations / 4000 exhaustive random instances, worst LHS/RHS ratio 1.000000). Repair = cite Finner.
(3), §2.1	VALID	Immediate from (2) with $f_i = e^{\lambda(Y_i - EY_i)}$ > 0 .
(4), §2.2 (spacing moments)	VALID	$B_r \sim \text{Beta}(1, r)$, $E B_r^j = j! \Gamma(r+1) / \Gamma(r+j+1) = j! / ((r+1) \cdots (r+j))$. Re-derived and checked numerically (max error $1.5e-11$). The remark “including the two endpoint spacings” matters and is correct — this is where an off-by-one would live, and there is none: r interior points give $r+1$ exchangeable spacings, each $\text{Beta}(1, r)$. $r \geq d/4 \implies E(dB_r)^j \leq 4^j j!$; $\log(1+x) \leq x$; $\sum_{j \geq 2} (4 \theta)^j \leq 32\theta^2$ for $ \theta \leq 1/8$. Verified to 60-digit precision by the series $E e^{sB_r} = \sum_j s^j / \prod_{i \leq j} (r+i)$ over $d \in \{4, \dots, 10^5\}$, all admissible r , and 81 values of $\theta \in [-1/8, 1/8]$: 0 violations, max(lhs-rhs) = 0 (equality only at $\theta=0$).
(5), §2.2 (mgf bound)	VALID	$X_v = (C_v + U_v)/2$ is $U[0, 1]$; low–low sums ≤ 1 with equality a.s. impossible, high–high sums ≥ 1 always, cross sums $\geq 1 \iff U_u + U_v \geq 1$. No boundary off-by-one : the three cases are exhaustive and the cross case is an exact $\iff$. Brute-forced on 3148 vertices over 300 random regular graphs: 0 violations. Given C and $U_v = u$, $\deg_H(v) - b_v \sim \text{Bin}(r_v, u)$; $\int_0^1 \theta^j C(r, j) u^j (1-u)^{r-j} du = 1/(r+1)$ for $0 \leq j \leq r$, else 0. Checked by quadrature for all $d \in \{3, 6, 9\}$, both bits, all r , all k : max error $1.9e-11$.
(7)/(8), §3 ($q_{\{v, k\}}(C) = \Pr(\deg=k \mid C)$)	VALID	

step	label	note
(9), §3 ($\Pr(E^c) \leq n e^{-d/16}$)	VALID	$r_v \mid C_v \sim \text{Bin}(d, 1/2)$ (neighbours' bits are independent of C_v); multiplicative Chernoff with $\delta=1/2$ gives $\exp(-\delta^2(d/2)/2) = e^{-d/16}$. Exact binomial CDF check for $d = 1 \dots 3000$: 0 violations.
§4 conditional independence	VALID	This is the crux and it holds. For $v \in V_a$, $\deg_H(v)$ is a function of U_v and of U on $V_{\{1-a\}}$ only — the monochromatic edges are already absorbed into b_v . So conditioning on C and on $U_{\{V_{\{1-a\}}\}}$ makes $\{1\{\deg_H(v)=k\}\}_{v \in V_a}$ independent. Checked structurally: re-randomising one $U_w, w \in V_a$, never changed $\deg_H(v)$ for another $v \in V_a$ (0 violations over 300 graphs $\times$ both sides, and 0 in the earlier script's variant).
(10), §4	VALID	$p_{\{v, k\}}$ is the $(k-b_v)$ -th spacing of the r_v points $1-U_u$, hence $\text{Beta}(1, r_v)$ marginally given C , with mean $q_{\{v, k\}}$; uniform spacings are exchangeable so the index does not matter, including the two endpoint spacings.
(11), §4	VALID	$1+\eta p \leq e^{\eta p}$ for all real ηp ; $\eta = e^{\lambda-1} > -1$ so every factor is positive. Both signs of λ are legitimately covered.
(12), §4	VALID	Each $U_u, u \in V_{\{1-a\}}$, is read by $ N(u) \cap V_a \leq d$ of the $p_{\{v, k\}}$, so (2) applies with $R = d$; then (5) with $\theta=\eta$, $r=r_v \geq d/4$ on E . The degenerate case $p_{\{v, k\}} \equiv 0$ (k out of range) is handled and contributes 0 to the mean term and $32\eta^2$ to the quadratic term, as claimed. Arithmetic re-derived: $(1/d)\sum_{v \in V_a} [\eta d q_{\{v, k\}} + 32\eta^2] = \eta Q_{\{a, k\}} + 32(V_a /d)\eta^2$.
(13), §4	VALID	$e^{\lambda-1-\lambda} \geq 0$ always and $\leq \lambda^2$ for $ \lambda \leq 0.7$; $Q_{\{a, k\}} \leq 4 V_a /d$ on E since $q_{\{v, k\}} \leq 1/(r_v+1) \leq 4/d$; $\eta^2 \leq 3\lambda^2$ for $ \lambda \leq 1$. Constants left implicit but they exist and are absolute. The constraint $ \eta \leq 1/8$ forces $\lambda_0 \approx 0.117$; absolute, as claimed.
(14), §4 (Cauchy–Schwarz across the two sides)	VALID	Needs (13) at 2λ , i.e. $ \lambda \leq \lambda_0/2$; $ V_0 + V_1 = n$. This is the right way to handle the (correctly noted) dependence between the two sides.
(15), §4	VALID	Standard sub-gamma Chernoff from log-mgf $\leq C_1(n/d)\lambda^2$ on $ \lambda \leq \lambda_1$: optimising gives $\exp(-dt^2/(4C_1 n))$ in the Gaussian branch and $\exp(-\lambda_1 t/2)$ beyond, i.e. $\exp(-c \min\{dt^2/n, t\})$. Holds uniformly over every $C \in E$, as asserted — the bound depends on C only through $ V_a \leq n$ and $r_v \geq d/4$. Covers $k = 0$ and $k = d$.
(16), §5	VALID	$\tilde{q}_{\{v, k\}} \leq 1/(r_v+1) \leq 4/d$ on $\{r_v \geq d/4\}$, and ≤ 1 otherwise, so $\leq 4/d$ in all cases only because the truncation kills the other branch — correct as written. $\tilde{q}_{\{v, k\}}$ is measurable w.r.t. the bits in $N_G[v]$ (b_v , r_v , and the truncation indicator all are). Each bit lies in exactly $d+1$ closed neighbourhoods (G is d -regular), so (3) with $R = d+1$ + Hoeffding's lemma on $[0, 4/d]$: $(1/(d+1)) \cdot n \cdot \lambda^2 (d+1)^2 (16/d^2)/8 \leq 2n\lambda^2(d+1)/d^2 \leq 4n\lambda^2/d$. Sub-Gaussian tail $2\exp(-dt^2/(16n))$. Re-derived; the $(d+1)/d \leq 2$ slack is real.
(17), §5	VALID	

step	label	note
(18), §5	VALID	$\Pr(\deg_H(v)=k) = \int_0^1 C(d,k)x^k(1-x)^{d-k}dx = 1/(d+1)$ exactly, hence $E m_k = E Q_k = \mu$. This matches FLP/Alon–Wei (their Theorem 2.1 discussion) and Montgomery–Pokrovskiy–Sudakov Theorem 2.1. Verified exactly for d up to 33.
(19), §5	VALID	$0 \leq \mu - E \bar{Q}_k = \Sigma_v E[q_{\{v,k\}}1_{\{r_v < d/4\}}] \leq n e^{-d/16}$ using $q \leq 1$ and (9).
(20), §5	VALID	Trivially, the truncation is inactive on E .
(21), §6	VALID	Triangle inequality $ m_k - \mu \leq m_k - Q_k + \bar{Q}_k - E\bar{Q}_k + E\bar{Q}_k - \mu $ on E ; union over $d+1$ values of k for the two stochastic terms; E^c charged once. The double use of $B = n e^{-d/16}$ as both an additive deviation and a probability is a pun but both bounds are individually correct.
(1) from (21), §6	VALID	$d \geq 64 \log n \implies B \leq n \cdot n^{-4} = n^{-3}$; $t = \sqrt{(n/d) \log n} + \sqrt{\log n}$ gives $dt^2/n \geq A^2 \log n$ and $t \geq A \log n$, so $4(d+1)e^{-cA \log n} \leq 4n^{1-cA} \rightarrow 0$ for A large.
(22), §6	VALID	$t/\mu = \sqrt{(d \log n/n) + d \log n/n} = \sqrt{o(1)}$ exactly under $d = o(n/\log n)$.
§7 (23)	VALID	$I_{\{v,k\}}$ reads $N_G[v]$, read- $(d+1)$, $E I_{\{v,k\}} = 1/(d+1) = p$ by (18); (2) gives $E e^{\lambda m_k} \leq (1-p+pe^{\lambda R})^{n/R}$ with $R = d+1$, which is the mgf of $R \cdot \text{Bin}(n/R, p)$ with mean $np = \mu$; the Chernoff bound follows from the mgf alone, so $2\exp(-\epsilon^2 n / (3(d+1)^2))$ is correct for $0 < \epsilon \leq 1$, both tails.
§7 (24) and the choice ϵ_n	VALID	$\epsilon_n = 4(d+1)\sqrt{(\log n/n)}$ gives exponent $16 \log n / 3 > 5 \log n$, so $2(d+1)n^{-5} \rightarrow 0$; and $\epsilon_n \leq \sqrt{256 \log n / \sqrt{(\log n/n)}} = \sqrt{o(1)}$ throughout $d < 64 \log n$. $d = 0$ handled separately ($m_0 = n = \mu$).
§7 coverage	VALID	§6 covers $d \geq 64 \log n$, §7 covers $d < 64 \log n$; together every n in a sequence with $d = o(n/\log n)$, including sequences alternating between the regimes, since both bounds are uniform.
§8	VALID	Accurate self-description; correctly identifies that the gain over the earlier attempt is that the <i>random</i> part of the model is bipartite after bit exposure, regardless of $\chi_f(G)$.

No step is ERROR. The single GAP is the justification (not the truth) of Lemma (2).

B.3 Reference check

The writeup is self-contained and cites nothing explicitly, so there is no misquoted citation to catch. What I did check:

1. **arXiv:2207.13651 (Fox–Luo–Pham)** — fetched the PDF and read it. Confirmed: Theorem 1.3 is $d = o(n/(\log n)^{12})$; the conjecture sentence is exactly the target; Conjecture 1.2 (Alon–Wei, asymptotic) and Conjecture 1.1 (exact, ± 2) are as the catalog says. Confirmed the crux stated in my brief: FLP’s proof is a vertex-exposure martingale plus Freedman/Fan–Grama–Liu Bernstein (their Lemma 3.1), with $|Y_i| \leq C \log n$ (Lemma 3.2) and predictable quadratic variation $M_n \leq C(\log n)^{11} n/d$ (Lemma 3.3). One log from $|Y_i|$, eleven from M_n — hence $(\log n)^{12}$. The truth for M_n is n/d , and the writeup’s three pieces each have variance proxy exactly n/d , so the writeup is quantitatively consistent with FLP’s own heuristic. I also checked that FLP’s interval-indicator conditioning (their σ -algebra G in the proof of Lemma 3.2) is *local and only inside the martingale-difference bound*; it is not the global half-interval decoupling used here. The idea in §3 does not appear in FLP.
2. **Lemma (2)** — confirmed against Finner, *A generalization of Hölder’s inequality and some probability inequalities*, Ann. Probab. 20 (1992) 1893–1901: for a product probability space and f_i depending on coordinates S_i , if $\sum_{i: j \in S_i} c_i \leq 1$ for each j , then $\int \prod f_i^{c_i} \leq \prod (\int f_i^{c_i})$. Taking $c_i = 1/R$ and applying to f_i^R is exactly (2). **Missing attribution is the one reference defect.**
3. **Hoeffding’s lemma** (used unnamed in §5) and the multiplicative Chernoff bound (§7, and for (9)) are standard and correctly applied.

Priority check (step 2b) — searched, and the target is NOT already resolved, but the context has changed materially in 2026:

- **arXiv:2606.30612**, Montgomery, Pokrovskiy, Sudakov, *Nearly-uniform degree distributions in spanning subgraphs* (29 Jun 2026) — **proves Alon–Wei Conjecture 1.2 in the full range $d = o(n)$** , i.e. every d -regular graph *contains* a spanning subgraph with $(1 \pm \epsilon)n/(d+1)$ vertices of each degree. I fetched and read the relevant sections. This is a *different* statement: they take a typical irregular random subgraph $H_\emptyset$ and then *modify* it with a correction function ψ , and they state explicitly (their §1 and §2) that “as noted by Fox, Luo and Pham, the irregular random subgraph itself is not expected to satisfy (1) in the whole range $d = o(n)$ ”, and they cite $d = o(n/(\log n)^{12})$ as the state of the art **for the random subgraph itself**. The properties of $H_\emptyset$ they establish (their A1, A2 in Lemma 3.1) are aggregated (prefix sums, and blocks of length $\geq d^{\{2/3\}/3}$), explicitly *not* per-degree counts. So MPS does **not** imply the target.
- **arXiv:2607.06465**, *Irregular subgraph in a regular graph* (7 Jul 2026) — resolves Conjecture 1.1 exactly for fixed d and large n ; also cites FLP’s $(\log n)^{12}$ as the state of the art for the random model. Does not imply the target.
- Searches over the 2026 math.CO listing and the citing literature of FLP turned up no paper improving the *random model’s* range beyond $o(n/(\log n)^{12})$.

Self-contamination check (step 2c). None of the above rests on a search-engine paraphrase. I downloaded and read the actual PDFs of arXiv:2207.13651 (Fox, Luo, Pham, 27 Jul 2022, RSA 64 (2024)), arXiv:2606.30612 (Montgomery, Pokrovskiy, Sudakov, 29 Jun 2026) and arXiv:2607.06465 (7 Jul 2026), and quoted from their own text; all three are independent papers with named authors, dates and arXiv identifiers. Nothing I used as prior art traces back to this repository or its pull request, and I am not recording ALREADY_KNOWN in any case.

Net: the theorem proved here is, as far as I can determine, new; the **motivating** conjecture (Alon–Wei 1.2) it was designed to serve has meanwhile been settled in a stronger range by other means, which reduces the result’s importance but not its correctness or novelty. The writeup’s own caveat (“literature priority has not been independently checked”) is therefore discharged: not already known, but the surrounding landscape moved three months before the attack ran, and any writeup should now cite MPS.

B.4 Computational check

Scripts (reproducible) in `verification_astra/scripts/2207.13651__00/`: `referee_fast.py` (new, the main battery), `check_eq5.py` (new, high-precision), plus the surviving `check_core.py` / `check_concentration.py` from the killed run, whose results I re-derived rather than trusted. Note for reproducibility: `check_core.py` is slow (dense `linspace` quadrature inside nested loops) and was killed again by a timeout on this run without producing output; it is superseded by `referee_fast.py` (same checks A–G, faster) and `check_eq5.py` (exact series in place of its quadrature), both of which ran to completion. `check_concentration.py` ran to completion. Exact numbers obtained:

- **eq. (6) decomposition** — 3148 vertices over 300 random regular graphs ($d \leq 6, n \leq 16$): **0 violations**.
- **eq. (7) $q_{\{v,k\}} = \Pr(\deg=k | C)$** — exact quadrature, $d \in \{3, 6, 9\}$, both bits, all $r \in [0, d]$, all k : **max error 1.9e-11**.
- **E_C $q_{\{v,k\}}$ exactly** — $d \in \{1, 2, 3, 5, 8, 13, 20, 33\}$: **max error 1.4e-17** (floating point zero). Per-clique version ($K_{\{d+1\}}$, $d \in \{3, 6, 11\}$): every k gives exactly 1.0000.
- **§4 conditional independence** — perturbing one U_w , $w \in V_a$, never changed $\deg_H(v)$ for $v \in V_a \setminus \{w\}$: **0 violations** over 300 graphs $\times$ both sides (and 0 in the independent earlier implementation).
- **eq. (5) mgf bound** — mpmath, 60 digits, exact series for $E e^{\lambda \{sB_r\}}$; $d \in \{4, 8, 16, 32, 64, 128, 512, 2048, 8192, 10^5\}$, $r \in \{[d/4], \dots, d\}$, 81 values of $\theta \in [-0.125, 0.125]$: **0 violations**, **max(lhs - rhs) = 0** (equality only at $\theta=0$). (An earlier float version reported 14 “violations” of size $\sim 1e-8$ at $\theta=0$ plus overflow at $d=8192$ — pure quadrature artefacts, eliminated by the exact series.)
- **eq. (4) spacing moments** — $r \leq 29, j \leq 5$: **max error 1.5e-11**.
- **eq. (9) binomial tail** — exact `scipy` CDF, $d = 1..3000$: **0 violations**.
- **eq. (2) Finner / read-R Hölder** — 4000 exhaustive instances on small product spaces (random A_i , random positive tensors, $R = \text{true max multiplicity}$): **0 violations**, worst LHS/RHS = 1.000000.
- **Scale of the two stochastic pieces** (this is the quantitative heart of the claim): on $30 \times K_{10}$ ($n=300, d=9$, the adversarial high- χ_f case that defeated the earlier attempt), 3000 samples, $k=4$: $\text{Var}(m_k - Q_k | C) = 32.64$ vs the predicted $n/d = 33.3$ — **a match to 2%**; $E[m_k - Q_k] = -0.05$ (should be 0); $\text{Var}(Q_k)$ ranged over k in $[12.7, 16.5]$, comfortably under the bound $8n/d = 267$; $\text{Var}(m_k) \in [44.6, 64.3]$, consistent with $\leq 2\text{Var}(Q) + 2\text{Var}(m-Q)$ and with FLP’s $17n/(d+1) = 510$. On $21 \times K_{20}$ ($n=420, d=19$): $\text{Var}(Q_k) \in [4.8, 7.6]$ vs bound $8n/d = 177$. On a random 10-regular graph $n=300$: $\text{Var}(Q_k) \in [2.4, 10.2]$ vs bound 240. On $K_{\{100, 100\}}$: mean $Q_k \in [1.96, 1.99]$ vs $\mu = 1.98$, $\text{Var}(Q_k) \leq 0.195$ vs bound 16. I also hand-computed the clique case analytically (per clique: $\text{Var} \approx 2$, hence $\approx 2n/d$ in total, against the writeup’s proxy $\approx 4C_\emptyset n/d$) and it agrees with the simulation — the read-d Hölder bound is tight up to an absolute constant on exactly the family that killed the fractional-colouring approach.
- Empirical mgf check of (14) at $\lambda = \pm 0.04, \pm 0.08$ on $30 \times K_{10}$: satisfied with large margin.

Nothing failed. No finite object is load-bearing in this writeup (the proof is purely analytic), so there was no single construction to falsify; instead I verified every identity and inequality it rests on.

B.5 Caveats

1. **Lemma 2.1 has no proof and no citation.** The stated inequality is Finner’s (1992); the writeup’s sketch is wrong in wording (“adding a constant factor when $q < R$ ”) and would not survive a referee. This is the one thing a journal version must fix. It is a one-line repair (cite Finner, or Pelekis–Ramon’s read-k Hölder survey).
2. Constants in (13), (14), (15) are left as “absolute $C_\emptyset, C_1, \lambda_\emptyset, \lambda_1$ ” with $e^{\lambda-1-\lambda} = O(\lambda^2)$ waved through. They do exist (I instantiated them: $|\eta| \leq 1/8$ forces $\lambda_\emptyset \approx 0.117, \lambda_1 \approx 0.058$ after Cauchy–Schwarz, $C_\emptyset \approx 100$), but nothing is spelled out.
3. $B = n e^{\{-d/16\}}$ is used both as an additive deviation in (21) and as a probability. Both are legitimate but the equation reads as a pun; a careful version would name them separately.
4. The $d \geq 64 \log n / d < 64 \log n$ split is exhaustive, but §6 is only *useful* when $d \gg \log n$ — the writeup says so. §7’s ϵ_n is $o(1)$ only because $d < 64 \log n$ there; the two regimes must be read together and they are.
5. Edge cases: $d = 0$ is handled explicitly; $k = 0$ and $k = d$ are handled (the endpoint spacings, §2.2, are explicitly included — the classic off-by-one trap here, and it is closed); G disconnected or with multiple components is irrelevant since nothing beyond d -regularity is used; graphs are finite simple undirected, as the verdict caveat says. The measure-zero event $x(u) + x(v) = 1$ is correctly dismissed.

6. Nothing in the argument requires G to be connected, vertex-transitive, or of any particular chromatic number — the uniformity claim is genuine and is exactly the point.
7. **Contextual caveat the writeup could not know:** Montgomery–Pokrovskiy–Sudakov (arXiv:2606.30612, June 2026) settled the Alon–Wei asymptotic conjecture for all $d = o(n)$ by *modifying* the random subgraph. The present theorem is about the unmodified random subgraph and remains open in the literature, but it is now of narrower interest than it was when FLP posed it. The would_publish: true flag should be read in that light.
8. I could not find any error, but I want to record the residual risk honestly: this is a short elementary argument that removes eleven logarithms from a Random Structures & Algorithms paper. I checked every inequality by re-derivation and every identity numerically, and the one step that carries all the power (the conditional independence of each side of the random cut, §4) is both structurally verified by brute force and, on reflection, simply true — after exposing $1\{x(v) \geq 1/2\}$, monochromatic edges are deterministic, so the surviving randomness lives on a bipartite graph. That FLP missed it is plausible: their whole paper is committed to a martingale route, and their own use of interval indicators is local to Lemma 3.2.

B.6 Referee summary

I set out to break this and could not. The writeup’s idea is a single clean move: write $x(v) = (C_v + U_v)/2$ with C_v a fair bit, so that after exposing the bits every low–low edge is absent, every high–high edge is present, and only the cut edges remain random — which makes the *random* part of the model bipartite for **every** G , with no colouring and hence no $\chi_f(G)$ loss. Each side is then an independent set for the random part, so conditioning on one side’s labels makes the other side’s degree indicators independent Bernoulli with spacing-distributed parameters, and a read- d Finner/Hölder inequality plus the $\text{Beta}(1, r)$ mgf gives m_k a variance proxy of n/d around the random conditional mean $Q_k(C)$; $Q_k(C)$ is in turn a sum of $4/d$ -bounded read- $(d+1)$ functions of the bits, so it is sub-Gaussian with the *same* proxy n/d ; and $E Q_k = n/(d+1)$ exactly. Three pieces, each at scale $\sqrt{n/d}$, against a mean $n/(d+1)$ — which is $o(1)$ relative precision precisely when $d \log n = o(n)$, the conjectured threshold, with $d < 64 \log n$ mopped up separately by a read- $(d+1)$ Chernoff bound. Every step re-derived; every identity and inequality independently verified numerically, including the two that a decomposition argument usually dies on (the boundary case of the half-interval split, which is an exact $\iff$ with no off-by-one, and the endpoint spacings at $k = 0, d$, which are explicitly included); the predicted $\text{Var}(m_k - Q_k \mid C) \approx n/d$ matches simulation on disjoint cliques to 2%, on exactly the family that defeated the previous attempt. Priority: not already known — the two 2026 papers that resolve the surrounding Alon–Wei conjecture (Montgomery–Pokrovskiy–Sudakov, arXiv:2606.30612; arXiv:2607.06465) both still cite $d = o(n/(\log n)^{12})$ as the state of the art for the random subgraph itself, and neither implies this statement. The verdict is MINOR_GAPS rather than CONFIRMED for one reason: the load-bearing Lemma 2.1 is asserted with a justification that is not a proof and with no attribution, when it is in fact Finner’s 1992 inequality; that is a citation repair, not a mathematical hole.

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